DS-A2 | Inference and Uncertainty

Report template — complete with results from your own executed notebook

1. Question, estimand, hypotheses

State the question, unit, estimand, H0/H1, alpha, and minimum meaningful effect. Explain the threshold choice.

Your results / notes: _____

2. Data-generating process

Describe distributions, sample sizes, noise, effects, seed, software versions, and assumptions.

Your results / notes: _____

3. Sampling distribution

Insert the generated figure. Report repetitions, estimate center/spread, and comparison with the true effect.

Your results / notes: _____

4. Confidence interval and test

Report effect estimate, CI method/bounds, p-value, and practical interpretation.

Your results / notes: _____

5. Power analysis

Insert figure/table. Explain changes with sample size, noise, and effect; state repetitions and Monte Carlo uncertainty.

Your results / notes: _____

6. Counterexample

State the assumption violation and compare baseline/counterexample results.

Your results / notes: _____

7. Limitations and counterevidence

Discuss synthetic-data scope, assumptions, finite repetitions, and results that challenged expectations.

Your results / notes: _____

8. Independent verification and AI use

Name independently checked results, second method, agreement/tolerance, and AI suggestions accepted/rejected.

Your results / notes: _____

9. Reproduction

Give setup/run instructions and repository URL plus commit/tag after verification.

Your results / notes: _____